$$B = \frac{2\pi}{\lambda} \int_L n_2 I_p(z)\, dz .$$

$$I(z) = I, \qquad B \propto n_2 I l$$

$$\begin{cases} z_f = \dfrac{0.367\, ka^2}{\left[\left(\sqrt{P/P_c} - 0.852\right)^2 - 0.021^2\right]^{1/2}} \propto \dfrac{1}{P/P_c - 0.852} \\ z_{fm} = \dfrac{0.367 ka^2}{0.002} \propto \sqrt{\dfrac{P_c}{P}} \end{cases}$$

$$\Rightarrow l = z_{fm} - z_f \propto \left[\frac{1}{0.002} - \frac{1}{P/P_c - 0.852}\right]$$

$$P_c \propto \frac{1}{n_2} .$$